

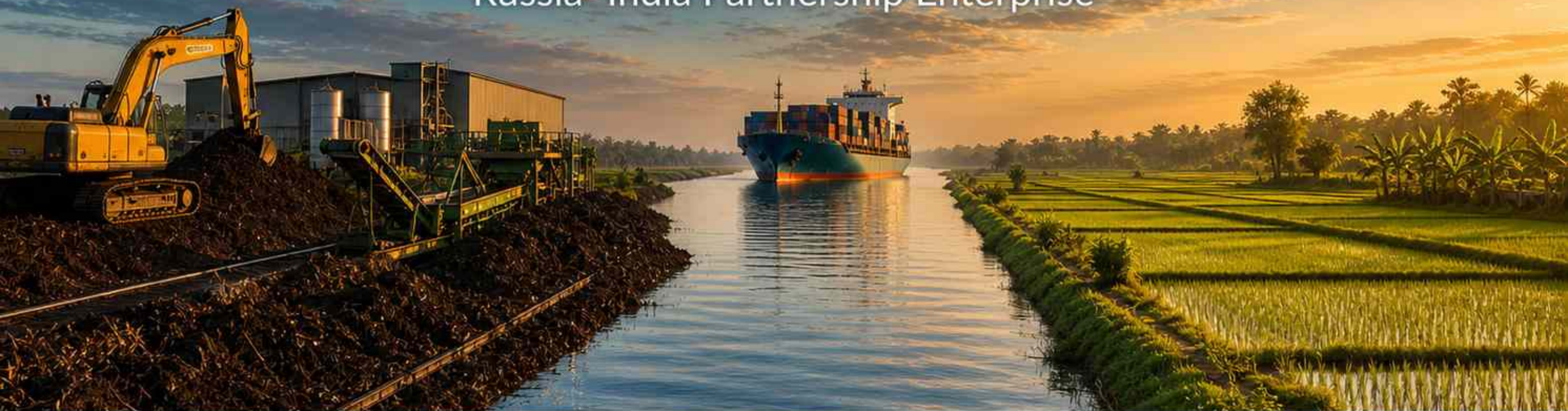
LOCALISATION OF LIQUID PEAT HUMIC FERTILISER PRODUCTION IN INDIA



BALTORGANIC



Russia-India Partnership Enterprise



*From Baltic peatlands
to Indian fields*



RUSSIA-INDIA PRODUCTION CHAIN



BALTORGANIC

Peat Humic Gel manufacturer



Peat extraction



Separation and cleaning



Reactor unit



Loading of IBCs



1-7

KINGISEPP, RUSSIA



9-18

INDIA

INDIA - FINISHED PRODUCTS



Peat Humic Gel products



Blending production



Finished products



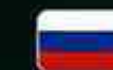
Fields under cultivation

RUSSIAN FEDERATION

- ① Extraction of lowland peat
- ② Primary preparation and cleaning
- ③ Transportation, storage, homogenisation of raw material
- ④ Mechanical separation and dosing
- ⑤ Preparation of peat pulp
- ⑥ Mechanochemical activation – Peat Humic Gel
- ⑦ Quality control, packaging (IBCs)
- ⑧ Sea transportation

INDIA

- ⑨ Receipt and storage of Peat Humic Gel
- ⑩ Homogenisation and stabilisation of Peat Humic Gel
- ⑪ Mechano-cavitation processing of Peat Humic Gel – humic concentrate
- ⑫ Technological adaptation of concentrate
- ⑬ Production of product line
- ⑭ Packaging of finished products
- ⑮ Logistics and distribution
- ⑯ On-farm application
- ⑰ Soil reclamation and restoration
- ⑱ Monitoring and performance evaluation



BALTORGANIC

RAW MATERIAL SUPPLY FOR PRODUCTION

Russian Federation • Raw material preparation, stages 1–6

1



Lowland peat extraction

2



Primary cleaning

3



Transportation to site

4



Storage and homogenisation

5



Mechanical separation

6



Accumulation and dosing



RAW MATERIAL PREPARATION CYCLE

The Russian part of the project begins with the raw material cycle — from lowland peat extraction to obtaining a homogeneous mass ready for processing. At each stage, quality is monitored, impurities are removed, and the composition is stabilised. This ensures uninterrupted and stable operation of the entire production line.



1 LOWLAND PEAT EXTRACTION

excavator method, layer-by-layer extraction, visual quality control



2 PRIMARY MECHANICAL CLEANING

removal of roots, wood residues and stones



3 TRANSPORTATION TO SITE

by trucks with batch control and incoming inspection



4 RECEIPT, STORAGE AND HOMOGENISATION

moisture and composition control



5 MECHANICAL SEPARATION

on trommel screens and vibratory screens, removal of heavy inclusions



6 ACCUMULATION AND DOSING

in bunker-hopper and metered peat feed to the line

PEAT HUMIC GEL PRODUCTION

Russian Federation • technological line



PULP

ACTIVATION & EXTRACTION

ACCUMULATION

CONTROL

DISPATCH



PEAT PROCESSING CYCLE

Prepared peat is converted into marketable Peat Humic Gel: peat pulp is prepared, mechanochemical activation and alkaline extraction are carried out in a reactor unit, the finished product is accumulated, homogenised and after laboratory quality control is dispatched.



1 PULP PREPARATION

in tanks with intensive mixing and dispersion



2 PREPARATION FOR PUMPING

protection of pumping equipment from abrasive inclusions



3 FEED TO REACTOR UNIT

by high-pressure screw pumps



4 ACTIVATION & EXTRACTION

mechanochemical activation of peat and alkaline extraction of humic substances



5 PEAT HUMIC GEL OBTAINING

with high content of humic and fulvic acids



6 ACCUMULATION & HOMOGENISATION

storage of finished Peat Humic Gel in buffer tanks



7 LABORATORY CONTROL

quality control and issuance of batch passports



8 PACKAGING IN IBCs

filling Peat Humic Gel into 1,000 L IBCs

PREPARATION AND INTERNATIONAL LOGISTICS

Russian Federation • export shipment, stages 1–6

1



Filling Peat Humic Gel into 1,000 L IBCs

2



Formation of export consignment

3



Loading into sea containers

6



Sea delivery to the Republic of India



EXPORT SHIPMENT CYCLE

Finished Peat Humic Gel leaves the Russian site in a transport-ready form: the product is packaged in IBCs, an export consignment is formed, loaded into sea containers at the Port of Novorossiysk and shipped to the Republic of India. Each consignment is accompanied by a quality passport and a complete set of export documents.



1 PACKAGING INTO IBCs

Peat Humic Gel is packaged in 1,000 L IBCs, labelling and paperwork for the consignment.



2 FORMATION OF EXPORT CONSIGNMENT

storage of filled IBCs and preparation of the consignment for shipment.



3 LOADING INTO SEA CONTAINERS

securing and sealing IBCs in containers.



4 DELIVERY TO PORT

road delivery of containers to the Port of Novorossiysk.



5 CUSTOMS CLEARANCE

customs procedures and loading onto container ships.



6 SEA TRANSPORTATION

sea delivery to the Republic of India.



Republic of India

PEAT HUMIC GEL RECEIVING AND PREPARATION

Republic of India • Peat Humic Gel receiving, stages 1–4



1 Receiving and inspection



2 Decanting and transfer



3 Production of humic concentrate



4 Recipe adaptation



RECEIVING AND PREPARATION CYCLE

The partner plant receives Peat Humic Gel, monitors the quality of each batch, decants the IBCs, and prepares it for production – adapting it to local crops, soils, and climate conditions.



1 RECEIVING AND INSPECTION

Receiving containers and incoming inspection of Peat Humic Gel batches.



2 DECANTING AND TRANSFER

Decanting of IBCs, transfer to receiving tanks.



3 PRODUCTION OF HUMIC CONCENTRATE

Filtration and production of specialised products.



4 RECIPE ADAPTATION

Adaptation to the climate, soils, and crops of the Republic of India.

AGROBIOTECHNOLOGY CENTRE AND SOIL RECLAMATION

Republic of India • partner research centre



**COMPETENCY
CENTRE**



1



**Humic technologies
laboratory**



2



**Microbiology
laboratory**



3



Seed trials



4



Soil testing



5



**Specialist
training**



6



**Russia-India
joint programmes**



1 HUMIC TECHNOLOGIES

control of Peat Humic Gel
composition and humic acids.



2 MICROBIOLOGY

selection and control of beneficial
soil microflora.



3 SEED TRIALS

assessment of germination
and crop response.



4 SOIL TESTING

analysis of salinity, acidity
and fertility.



5 CAPACITY BUILDING

training of Indian agronomists
and technologists.



6 RUSSIA-INDIA PROGRAMMES

joint research and experience
exchange.

PRODUCTION OF SPECIALISED PRODUCTS

Republic of India • technological line, stages 1–6



RECEIVING

PROCESSING

FILTRATION

BLENDING

FILLING



PRODUCT PRODUCTION CYCLE

At the partner plant, Peat Humic Gel undergoes mechano-cavitation processing and filtration to obtain a humic concentrate, which according to the formulation is diluted in a base solution. The addition of mineral, biological, and microbiological additives creates a wide range of product lines for specific crops, soils, and agronomic challenges – a tailored solution for every client.



① PEAT HUMIC GEL RECEIVING TANKS



② MECHANO-CAVITATION PROCESSING OF PEAT HUMIC GEL



③ PEAT HUMIC GEL FILTRATION



④ FORMULATION AND BLENDING UNIT



⑤-⑥ FILLING SECTION FOR VARIOUS PACKAGING

PACKAGING AND LOGISTICS

Republic of India • product shipping, stages 1–4



 Packaging in consumer containers



 Packaging in IBCs



 Batch warehousing



 Delivery and shipping



PACKAGING AND SHIPPING CYCLE

Finished products are packaged across all segments – from retail to farms and agriholdings. Batches are warehoused, formed for shipping and delivered to agricultural enterprises and distributors throughout the Republic of India.



1 PACKAGING IN CONSUMER CONTAINERS

filling finished product into canisters and bottles for retail



2 FARM AND INDUSTRIAL PACKAGING

packaging in IBCs and large containers for farms and agriholdings



3 WAREHOUSING

formation of shipment batches in the warehouse



4 DELIVERY

delivery to agricultural enterprises and distributors in the Republic of India

IMPLEMENTATION OF TECHNOLOGIES IN AGRICULTURE

Republic of India • product application, stages 1–4



1

 Seed processing



2

 Application during vegetation



3

 Fertigation and drip irrigation



4

 Ground and aerial application



PRODUCT APPLICATION CYCLE

The product line is used at all stages of agrotechnology – from seed treatment to foliar feeding, through fertigation, drip irrigation, and ground and aerial systems, increasing yields of crops of the Republic of India.



1 SEED TREATMENT

use of the product during pre-sowing seed treatment



2 APPLICATION DURING VEGETATION

foliar feeding of crops during the vegetation period



3 FERTIGATION AND DRIP IRRIGATION

application through irrigation systems



4 GROUND AND AERIAL APPLICATION

spraying with ground machinery and drones

CROP-SPECIFIC APPLICATION PROTOCOLS

Republic of India • field protocols, crops 1-5

This DIY liquid humate behaves similarly to commercial **12% liquid humic products**. Because Indian soils are often low in organic carbon, it acts as a "heavy lifter" to unlock fertilizers bound in the soil profile.

1 RICE (PADDY)

TARGET STAGE: Seedling transplanting, tillering, and panicle initiation.

- **Root Dip:** Mix 2 ml of liquid per liter of water. Dip the roots of rice seedlings for 30 minutes before transplanting.
- **Soil Drench / Main Field:** Apply 1 Liter per acre via flood irrigation channels during the tillering stage.

2 WHEAT

TARGET STAGE: Crown root initiation (CRI) and tillering stages.

- **Foliar Spray:** Mix 2-3 ml per liter of water and spray evenly over the foliage.
- **Irrigation:** Feed 1 Liter per acre alongside the first or second canal irrigation cycle.

3 SUGARCANE

TARGET STAGE: Sett planting and early tillering (45-60 days).

- **Sett Treatment:** Soak sugarcane setts in a 0.5% humate solution (5 ml/Liter) for 15-20 minutes before planting.
- **Drip / Fertigation:** 1.5 to 2 Liters per acre distributed across early growth flushes to promote thick tillers and rooting depth.

4 VEGETABLES (TOMATO, CHILI, ONION, POTATO)

TARGET STAGE: Vegetative growth and early flowering / tuber initiation.

- **Foliar Spray:** 2 ml per liter of water sprayed every 15 days. Helps prevent flower drop in chilies and tomatoes.
- **Drip Irrigation:** 500 ml to 1 Liter per acre every 20 days. This dramatically improves bulb / tuber sizing in onions and potatoes.

5 FRUIT ORCHARDS (MANGO, CITRUS, BANANA)

TARGET STAGE: Pre-flowering and post-fruit set.

- **Soil Basin Drenching:** Dilute 3 ml of liquid humate per liter of water. Apply 2 to 5 Liters of this diluted mix directly inside the basin of each mature tree. Repeat after fruit set to maximize size and weight.



GENERAL DOSAGE SUMMARY

Three application methods cover the whole season — dosage per litre of working solution or per acre of field, with the practice that keeps the product working.



1 FOLIAR SPRAY

2 ml per Litre of water.

Spray in early mornings or late evenings.



2 DRIP / FERTIGATION

1 Litre per Acre.

Mix directly into your main venturi fertilizer tank.



3 SEED / ROOT TREATMENT

2 ml per Litre of water.

Soak seeds for 4-6 hours or dip roots for 15 mins.

CRUCIAL IMPLEMENTATION TIPS

Fertilizer Savings. Combining liquid peat humate with commercial synthetic options like Urea or DAP improves nitrogen and phosphorus efficiency. In typical Indian farming systems, incorporating humates allows you to reduce traditional chemical fertilizer usage by **20-25%** without suffering yield losses.

POPULAR HUMIC-NPK BLENDS AND RATIOS

Republic of India • fertigation and foliar application

1 HUMIC + UREA / UAN SOLUTIONS

Blending liquid humic substances with nitrogen sources **reduces nitrogen volatilization and leaching**, enhancing leaf growth in crops like wheat and rice.

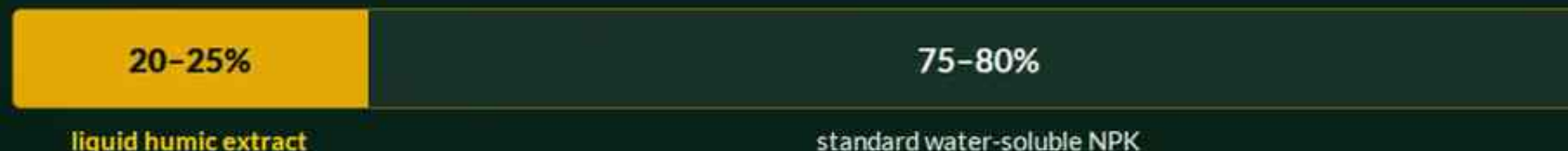
2 HUMIC + DAP (DIAMMONIUM PHOSPHATE)

Used heavily during sowing or early root development; humic acid **prevents phosphorus from fixing with calcium or iron** in Indian alkaline soils, keeping P available.

3 BALANCED LIQUID NPK + LIQUID HUMATE

For example **19:19:19** or **12:61:00**. Applied via drip fertigation to maximize root-zone uptake and minimize chemical fertilizer loss.

TYPICAL DRIP FERTIGATION RATIO



+20-30%

NPK absorption capacity

up to -25%

chemical fertilizer use (Urea, DAP)



KEY BENEFITS IN INDIAN FARMING

Liquid peat humic extracts are commonly blended with NPK (Nitrogen, Phosphorus, Potassium) fertilizers in Indian agriculture to boost nutrient use efficiency in depleted or nutrient-locked soils.



1 NUTRIENT EFFICIENCY

Increases NPK absorption capacity by **20% to 30%**, allowing farmers to cut down chemical fertilizer use (like Urea and DAP) by up to **25%**.



2 ROOT PROLIFERATION

Stimulates lateral root growth and enhances micro-flora activity in diverse agro-climatic zones.



3 APPLICATION MODE

Ideal for drip irrigation (fertigation) or foliar sprays during critical vegetative and flowering stages.

LAND RECLAMATION AND AGRONOMIC SUPPORT

Republic of India • land reclamation and support, stages 1–5

1



Restoration of degraded lands

2



Restoration of saline soils

3



Soil organic matter and bioactivity

4



Agonomic monitoring



LAND RECLAMATION AND SUPPORT CYCLE

Russia–India Partnership Enterprise technologies restore degraded and salinised lands, increase soil organic matter and biological activity. Effectiveness is confirmed by agronomic monitoring and economic evaluation.



1 RESTORATION OF DEGRADED LANDS

land restoration programmes



2 RESTORATION OF SALINE SOILS

soil desalination and restoration programmes



3 SOIL ORGANIC MATTER AND BIOACTIVITY

increase in organic matter and microbiota



4 AGRONOMIC MONITORING

effectiveness monitoring of application



5 ECONOMIC EFFICIENCY

assessment of return on technology implementation

DIVISION OF RESPONSIBILITIES

Russia-India Partnership Enterprise



RUSSIAN FEDERATION PRODUCTION

- 1 Lowland peat extraction
- 2 Primary purification and separation
- 3 Homogenisation and raw material preparation
- 4 Pulp preparation
- 5 Mechanochemical activation
- 6 Alkaline extraction
- 7 Peat Humic Gel production
- 8 Laboratory testing
- 9 Packaging in IBCs and logistics



REPUBLIC OF INDIA RECEIVING, PROCESSING, DISTRIBUTION

- 1 Receiving and incoming inspection
- 2 Humic concentrate production
- 3 Homogenisation and stabilisation
- 4 Adaptation to climate and soils
- 5 Recipe development
- 6 Product manufacturing
- 7 Packaging and warehousing
- 8 Field application
- 9 Land reclamation

From Baltic peatlands to Indian fields

LOGIC AND BENEFITS OF DIVISION

Why Peat Humic Gel is more profitable than dry peat

DRY PEAT IN BIG BALES



- ✓ Large volume and transport weight
- ✓ Low concentration of substances
- ✓ Requires full processing on-site
- ✓ Losses and dust during handling
- ✓ Higher costs per tonne of product

VS

PEAT HUMIC GEL IN IBCs



- ✓ Concentrate — lower volume and weight
- ✓ High content of humic acids
- ✓ Ready-to-use, fast startup
- ✓ Standard IBC packaging 1,000 L
- ✓ Lower delivery cost per tonne

From Baltic peatlands to Indian fields

JOINT RESULT

A partnership in which each side is irreplaceable



BALTORGANIC



BaltOrganic runs an active pilot-industrial development programme: peat processing technologies are refined on an operating line and verified by laboratory studies of the finished products. For us, these are not theoretical assumptions, but real working technologies delivering measurable results today.

BALTORGANIC CONTRIBUTES

- + Expertise and experience in peat processing
- + Proven production technologies
- + Stable source of quality raw materials
- + Laboratory testing and quality verification
- + Technology reliability guarantee



THE LOCAL PARTNER PROVIDES

- ✓ Pilot fields for agronomic adaptation of the product
- ✓ Production and logistics infrastructure
- ✓ Local scientific and human resources
- ✓ Access to markets and in-depth local expertise
- ✓ Support for implementation and solution promotion

By combining Russian technological depth with Indian operational strength, we not only enter the market — we set its standards.

The goal of the partnership is leadership in the humic products market in India and across Asia

Two countries — one chain, one result



NEXT STEP



ANALYTICS AND RECIPE

precise analysis of the market and priority crops, recipe map from the research team at Timiryazev Academy



BUDGET AND ROI

project budget with calculation of investment returns



PILOT BATCH

crop, volume, field for agronomic results

1

2

3

4

5

6



PRODUCTION LINE DESIGN

design of the partner's production line for priority products



ROADMAP

timelines and division of responsibilities



LEGAL STRUCTURE

joint venture shares and transfer pricing for the concentrate

**No concepts. No presentations.
A plan we can start working on tomorrow.**

We are ready to act. Right now.



Project Manager
Viktor E. Geshele



+7 921 918 38 88



v.geshele@mail.ru

PROJECT RESULTS



Technology that brings land back to life